

The empirical recurrence for OEIS A234339: recovering a conjecture that is not written down

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Abstract

OEIS A234339 records “Empirical recurrence of order 77”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 512$ vertices and satisfies some monic recurrence of order at most S ; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 77, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 77 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A234339 is “Number of $(n+1) \times (2+1)$ 0..7 arrays with every 2×2 subblock having the sum of the absolute values of the edge differences equal to 14.”. It has offset 1 and begins

9076, 411064, 15724414, 661471100, 27487124788, 1154009792030, 48437475144112, 2035786694647460, ...

The entry states:

Empirical recurrence of order 77 (see link above)

As of the “Last modified” line on the live entry (Jun 02 2025 09:17:01, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 512$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 512$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 1024$ exact terms of the model returns order 77 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{77} c_i a(n-i),$$

c_1	46	c_{21}	887710616704701640930	c_{41}	38761362519510759680396864617
c_2	686	c_{22}	−3326540630701289362292	c_{42}	−586525465202239901670293792807
c_3	−39131	c_{23}	−14205591683763300259754	c_{43}	−42211587509722646041687816792
c_4	−147608	c_{24}	56297099545629304041237	c_{44}	1260075442354729229942951363119
c_5	13161758	c_{25}	182735902915047707420017	c_{45}	166592726454454084372210872
c_6	5491556	c_{26}	−767978059616287745836565	c_{46}	−2182500915576822076327099787940
c_7	−2422071738	c_{27}	−1896332022514815683832791	c_{47}	96071521254690064489893640134
c_8	2669301671	c_{28}	8501727703353782268816913	c_{48}	3037262857751753136498131019873
c_9	280934675228	c_{29}	15888408497295948336347226	c_{49}	−194978093453624708331263307792
c_{10}	−544179239981	c_{30}	−76708964216142481329364980	c_{50}	−3384177795462660213307834500780
c_{11}	−22251655048299	c_{31}	−107260449257105303974342660	c_{51}	217011288989783962536339115686
c_{12}	54720382635220	c_{32}	565472654093678652002345121	c_{52}	3007012515559713352976779366246
c_{13}	1267432845547909	c_{33}	580358573439870006956132749	c_{53}	−143474728321082619553140285259
c_{14}	−3576653091034131	c_{34}	−3408463577123319957190881689	c_{54}	−2120011712535679103062495702038
c_{15}	−53817823438579862	c_{35}	−2492850609278114242335730243	c_{55}	40296428707275175849286144706
c_{16}	166796201636618054	c_{36}	16790757499970745612792206053	c_{56}	1177861084554219483602981121684
c_{17}	1748506550387989431	c_{37}	8363741285815559104050258937	c_{57}	21208736542570154113267479073
c_{18}	−5824255814758095441	c_{38}	−67493887881664764986109749144	c_{58}	−510855824006477751691754030283
c_{19}	−44303157179923285861	c_{39}	−21296199132742063284311047699	c_{59}	−30505073886353428786157245725
c_{20}	156893139869964306677	c_{40}	220851309477824621103121309649	c_{60}	170722777333502275178118799597

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 77, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $77 + 4$ terms removed returns order 77 as well, so $\deg q_{\min} = 77 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 13 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $77 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 77 that this sequence satisfies — and that family turns out to have exactly one member.

References

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